

I**CABLES**

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I.1. INTRODUCTION

The cable is an important and widely used structural element, in particular for suspension or cable-stayed bridges, cable-stayed pylons (optical cables in telecommunications), suspended covers (meeting room, sports hall, circus, etc.), support organs (cable cars, elevators), or attached to the hooks of lifting equipment (cranes, overhead cranes, etc.).

From a structural point of view, the cables are very efficient, because the material used develops only an internal tensile force under the effect of the load. No buckling can appear either under the effect of a bending moment or under the effect of a compressive force. The cable is even a competitive structural element. The steels used are very high strength, called cable steels.

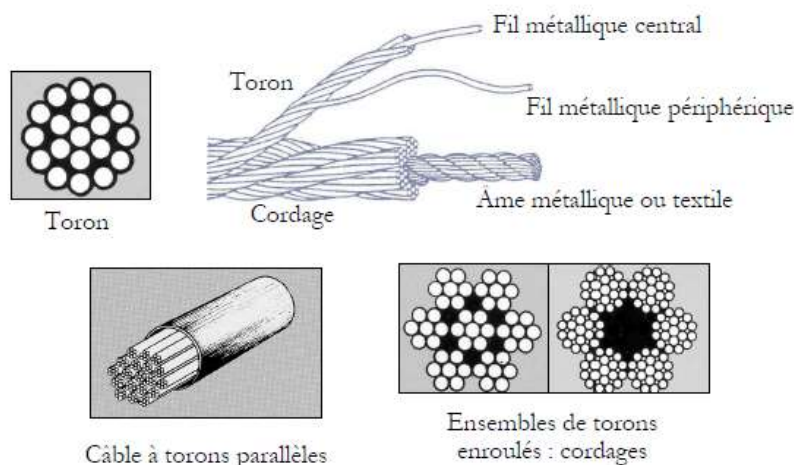


Figure (I.1): Cable constitution

I.2. ASSUMPTIONS: CHARACTERISTICS AND WORKING MODE OF A CABLE

The cable has a relatively small diameter compared to its length, making it a flexible element, or even perfectly flexible in specific cases. As a result, it offers minimal resistance to bending, torsion, shear forces, or compression. Consequently, the force acting within the cable always acts tangentially along its length and is referred to as 'cable tension.' Nevertheless, the cable structure exhibits a disadvantage: 'The deformed shape of the cable differs significantly from its initial shape, leading to important deformations.' Therefore:

1. The principle of linearity (in particular the principle of superposition) is not applicable.
2. The relations between the force in the cable and its slope are written in the deformed form (loaded cable).

We assume in this entire chapter that the cable is subjected to vertical loads only (horizontal forces are not considered here), hence:

✓ The horizontal component of the cable tension 'H' remains constant (see fig. I.2).

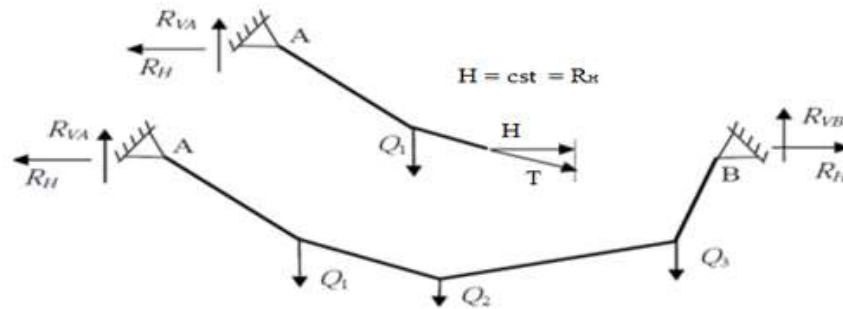


Figure (I.2): Horizontal component 'H' of the cable tension.

- ✓ The most inclined section is the one, which is subjected to the greatest tensile force, since H is constant, according to equation (I.1).

$$T = \frac{H}{\cos \alpha} \quad (\text{I.1})$$

- ✓ The bending moment is zero at any point of the cable.

I.3. CABLE SUBJECTED TO CONCENTRATED LOADS

A cable with negligible self-weight and subjected to concentrated loads at specific points on the cable, adopts a broken-line shape. This can occur in various situations, including:

1. Heavy lifting: When a cable is used in a lifting system, a concentrated load may be suspended using a hook, pulley, or other lifting device.
2. Hanging objects: When you hang a heavy object on a cable, the load is concentrated where the object is attached to the cable.
3. Entertainment equipment: In the entertainment industry, such as in theater productions, cables may be used to support lighting, sound, or set pieces. Loads can be concentrated at the attachment points of the cable to the equipment.

In these situations, ensuring that you design and size the cable to withstand the intended geometry and scenarios. The cable's geometry is maintained, however, if the cable deforms under the system, the system becomes statically determinate. In practice, two approaches are commonly encountered.

Figure (I.3): The effect of concentrated load on cable geometry

I.3.1. The geometry of the cable is known (Design approach)

In this case, the architect imposes the overall dimensions of the structure and asks the engineer to calculate the feasibility of his project (Fig.I.4a).

Data's: dimensions L , D , and charges (q value, Q_i + position L_i).

Variable: Internal forces and support reaction (pushing force), exact geometry (d_1 , d_2), length of deformed cable.

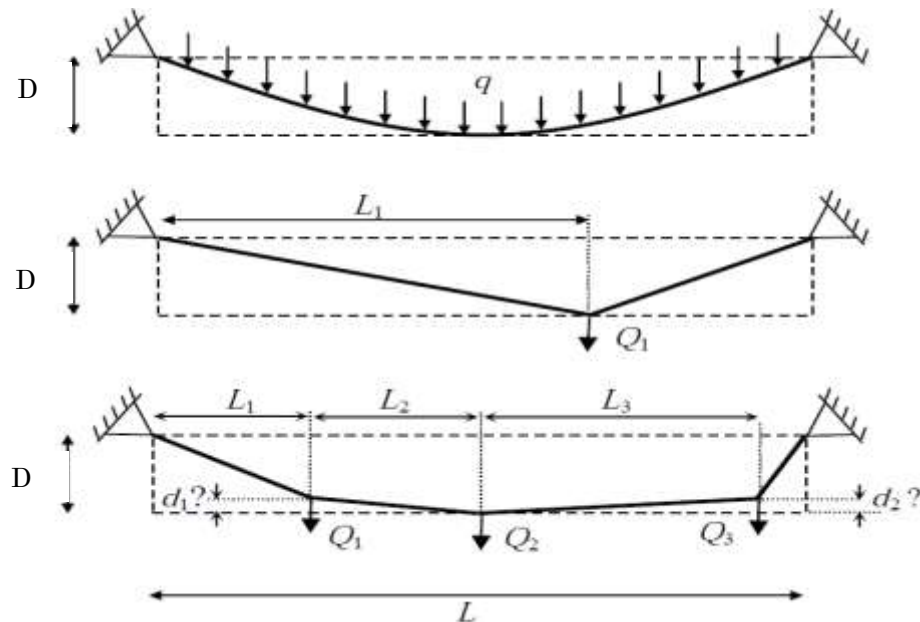


Figure (I.4a): Design approach

I.3.2. Geometry of the cable is unknown (expertise approach)

Two potential calculations may arise: one involving the problem of 'reaction (given data) – shape (unknown)' and the other involving the problem of 'length (given data) – reaction (unknown)'. In these scenarios (Fig I.4b), 'reaction' represents the compressive forces acting at the supports, commonly known as support reactions.

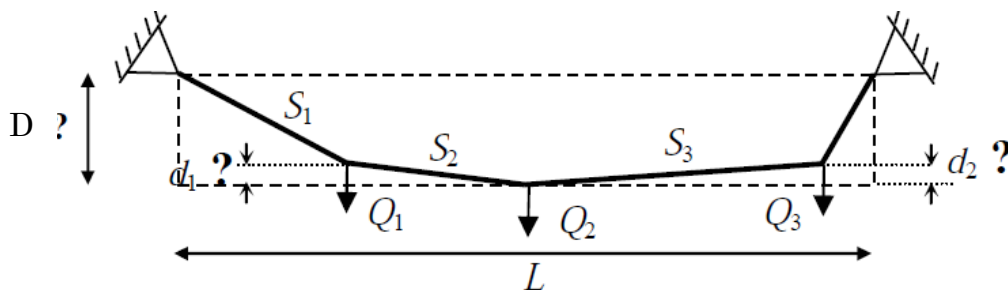


Figure (I.4b): 'Reaction-form', 'length-reaction' approaches.

I.3.3.Exemple

Determine the tension in each segment of the cable shown in Fig. I.5.

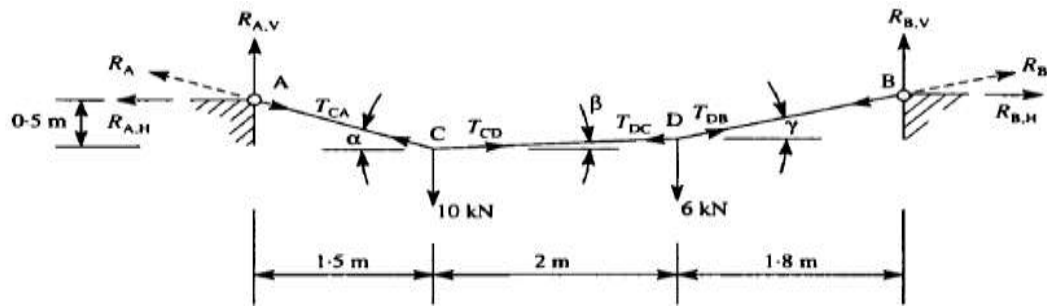


Figure (I.5): Example of a cable subjected to concentrated loads

Applying the equations of equilibrium yields

$$\sum M_B = 0$$

$$R_{AV} * 5.3 - 10 * 3.8 - 6 * 1.8 = 0$$

$$R_{AV} = 9.2 \text{ kN}$$

$$\sum M_C = 0$$

$$R_{AH} * 0.5 - R_{AV} * 1.5 = 0$$

$$H = R_{AH} = 27.6 \text{ kN}$$

Or:

$$R_{AV}/R_{AH} = \tan \alpha = \tan 18.4^\circ; R_{AH} = 27.6 \text{ kN}$$

We can obtain the tension in segment CA as follows:

$$T_{CA} = R_A = \sqrt{R_{AH}^2 + R_{AV}^2} = \sqrt{27.6^2 + 9.2^2} = 29.1 \text{ kN}$$

Now we can analyze the equilibrium of points C in the vertical direction

$$T_{CD} \sin \beta + T_{CA} \sin \alpha - 10 = 0$$

$$T_{CD} \sin \beta = 0.815 \quad (i)$$

Equilibrium of points C in the horizontal direction gives

$$T_{CD} \cos \beta - T_{CA} \cos \alpha = 0$$

$$T_{CD} \cos \beta = 27.612 \quad (ii)$$

Dividing equation (i) by (ii), we obtain:

$$\tan \beta = 0.0295, \quad \beta = 1.69^\circ, \quad T_{CD} = 27.6 \text{ kN}$$

I.4. CABLE SUBJECTED TO A UNIFORM HORIZONTALLY DISTRIBUTED LOAD

The cables can also support the weight of the beam and/or the deck of bridges with very large spans (Fig I.6). A suspension bridge is the typical example, wherein the deck is suspended by cables using a group of equally spaced hangers. The load funicular tells us directly about the deformation (geometry) of the cable (Fig I.6). The cable assumes a **parabola shape** in this case.

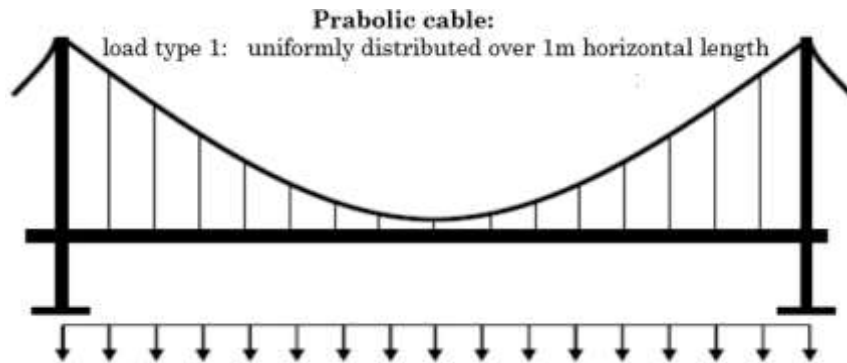


Figure (I.6): When a load is uniformly distributed over a 1-meter horizontal span (In the instance of the suspension bridge deck's weight).

In order to analyze this type of problem, we first determine the shape of the cable, subjected to a vertical load w_0 distributed uniformly on the horizontal (Fig. I.7a). The origin 0 of the axes (x, y) is located at the lowest point of the cable. We analyze a small section of length Δs of the cable (Fig. I.7b). The tension variation in this small segment of cable is denoted ΔT . The resultant of the charge is $w_0 \Delta x$ and acts at $\Delta x/2$ from the point 'O'.

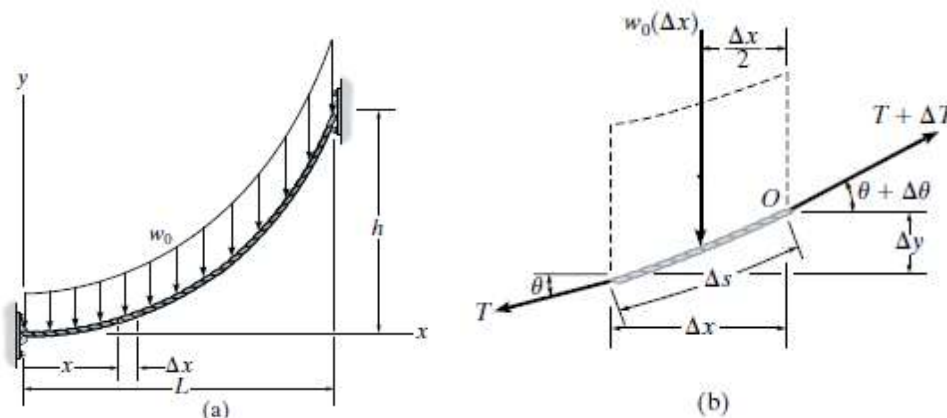


Figure (I.7): cable subjected to a uniform distributed load.

Applying the equations of equilibrium yields

$$\sum F_x = 0; -T \cos \theta + (T + \Delta T) \cos (\theta + \Delta \theta) = 0 \quad (\text{I.2a})$$

$$\sum F_y = 0; -T \sin \theta - w_0(\Delta x) + (T + \Delta T) \sin (\theta + \Delta \theta) = 0 \quad (\text{I.2b})$$

$$\sum M_{/O} = 0; w_0(\Delta x)(\Delta x/2) - T \cos \theta \Delta y + T \sin \theta \Delta x = 0 \quad (\text{I.2c})$$

Dividing each of these equations by Δx and taking the limit as $\Delta x \rightarrow 0$ and hence $\Delta y \rightarrow 0$, $\Delta \theta \rightarrow 0$ and $\Delta T \rightarrow 0$, we obtain:

$$\frac{d(T \cos \theta)}{dx} = 0 \quad (\text{I.3a})$$

$$\frac{d(T \sin \theta)}{dx} = w_0 \quad (\text{I.3b})$$

$$\frac{dy}{dx} = \tan \theta \quad (\text{I.3c})$$

Integrating equation (I.3a), where $x=0$, $T = H$, we have :

$$T \cos \theta = H \quad (\text{I.4})$$

Equation (I.4) implies that the horizontal component of the tension remains consistent at every point along the cable.

By integrating equation (I.3b), knowing that at $x=0$, $T \sin \theta = 0$, we have:

$$T \sin \theta = w_0 x \quad (\text{I.5})$$

By eliminating T through division of equation (I.5) by equation (I.4) and subsequently using equation (I.3c), we derive the slope of the cable at all points:

$$\tan \theta = \frac{dy}{dx} = \frac{w_0 x}{H} \quad (\text{I.6})$$

Let us integrate the equation above with $y=0$ to $x=0$:

$$y = \frac{w_0}{2H} x^2 \quad (\text{I.7a})$$

This is the equation of a parabola. The differential equation of the cable is then written as follows:

$$\frac{d^2 y}{dx^2} = \frac{w_0}{H} \quad (\text{I.7b})$$

The horizontal component H is obtained using the boundary condition $y=h$ at $x=L$. So:

$$H = \frac{w_0 L^2}{2h} \quad (\text{I.8})$$

Finally, substituting the equation below into equation (I.7a) gives:

$$y = \frac{h}{L^2} x^2 \quad (\text{I.9})$$

From equation (I.4), **the maximum tension in the cable occurs when the angle θ is maximum.**, i.e. at $x=L$. From equations (I.4) and (I.5), we have:

$$T_{max} = \sqrt{H^2 + (T \sin \theta)^2} \quad (\text{I.10a})$$

$$T_{max} = \sqrt{H^2 + (w_0 L)^2} \quad (\text{I.10b})$$

Using equation (I.8), we can express T_{max} as a function of w_0 :

$$T_{max} = w_0 L \sqrt{1 + \left(\frac{L}{2h}\right)^2} \quad (I.11)$$

Example: For the cable below (Figure I.8), calculate the maximum tension and the minimum tension.

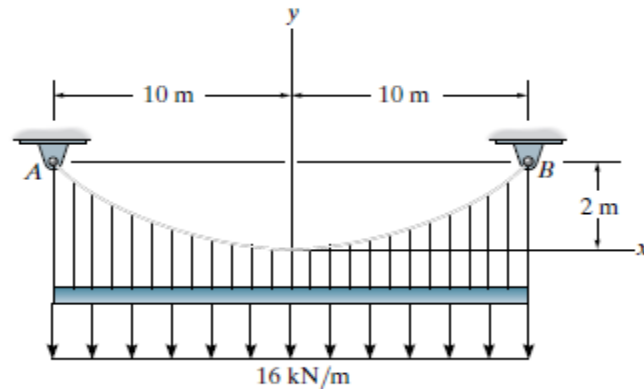


Figure (I.8): Symmetric cable subjected to a load uniformly distributed on the horizontal.

Using equation (I.11), we have:

$$T_{max} = w_0 L \sqrt{1 + \left(\frac{L}{2h}\right)^2} = 16 * 10 * \sqrt{1 + \left(\frac{10}{2 * 2}\right)^2} = 431 \text{ kN}$$

$$T_{min} = H = \frac{w_0 L^2}{2h} = \frac{16 * 10^2}{2 * 2} = 400 \text{ kN}$$

I.5. CABLE UNDER ITS OWN WEIGHT

In the preceding paragraph, it is important to note that we have disregarded the self-weight of the cable. The cable's weight is uniformly distributed load along its entire length and does not create a horizontal projection. When a cable is solely influenced by its own weight, it assumes a **chain-like shape** (Fig I.9a). However, as shown in (Fig I.9b), we can denote the following:

L: Length of the cable (span)

H: Deflection of the cable (sag)

For the majority of cable-supported structures with a low sag/span ratio i.e. high slenderness (L/H), this chain-like shape can be approximated by a parabolic curve. The assumption of important slenderness (L/H), which applies to most cables, allows us to treat loads of type 1 and 2 as equivalent, simplifying the adoption of a parabolic geometry.

The chain can be confused with the parabola shape from approximately $L/H=10$

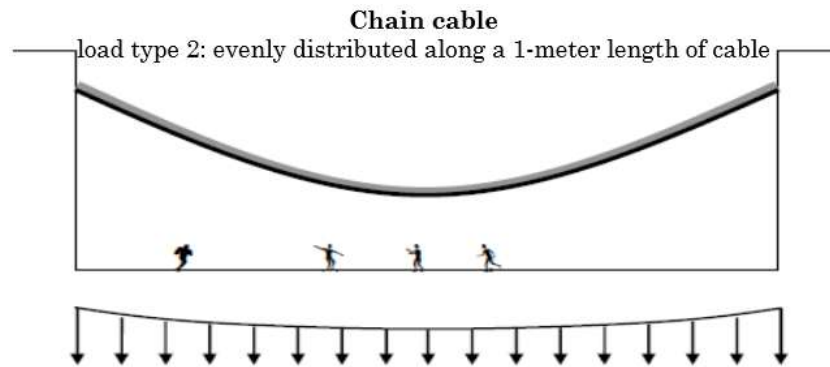


Figure (I.9a): When a load is distributed along a 1-meter length taken along the cable (such as the combined weight of the cable itself and a blanket hung directly on the cable), the load funicular forms a chain

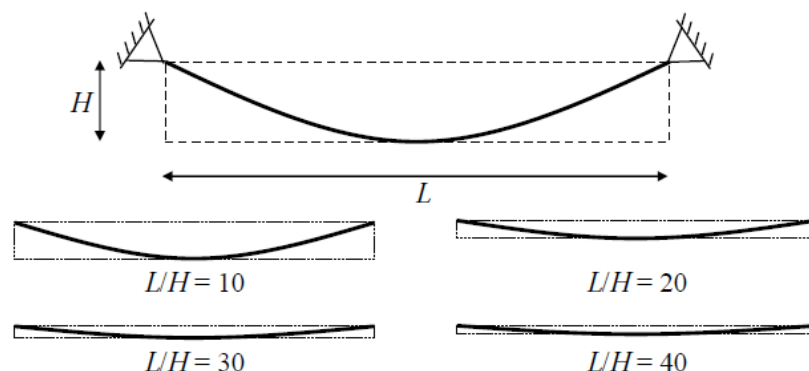


Figure (I.9b): The relationship between the cable's shape and its slenderness (L/H)

I.6. APPLICATION TO SUSPENDED BRIDGE

Suspension bridges are not only vital for spanning vast distances but also find application in shorter spans. The suspension bridge over the **'Dardanelles Strait in Istanbul'** holds the record for the longest central span, measuring an impressive 2023 meters, followed closely by **'Akashi Strait Bridge'** (also known as the Akashi-Kaykio Bridge) in Japan with 1991 meters central span. The fundamental concept behind suspension bridges involves supporting the deck's weight with two sturdy load-bearing cables, which are securely anchored either to the banks (in the case of self-anchored suspension bridges) or to external anchor blocks (in the case of externally anchored suspension bridges).

Assumptions: The own weight of the cable is very negligible compared to the weight of the deck. So, the cable is only subjected to a load distributed uniformly on the horizontal (weight of the deck), and then takes a parabolic shape Fig. (I.10).

The transmission of charges follows the following path:

- The weight of the deck is transmitted to the central cable via the hangers. The central cable is supported twice on the two piles; the static diagram of this suspension cable resembles that in Fig. (I.8).

- This cable is sealed to an anchor cable through a sealing system, which can be a pulley or a roller (simple support). This anchor cable transmits the tension of the main cable to an anchor block (foundation).
- The anchor block in turn transmits the forces to a good ground. The sealing angle (anchoring angle), ' β ' in Fig. I.10 is directly linked to the position of the good ground.
- It is known that the major disadvantage of the suspended bridge is that it requires the presence of an anchoring mass (and/or a rock) to retain the considerable forces exerted.

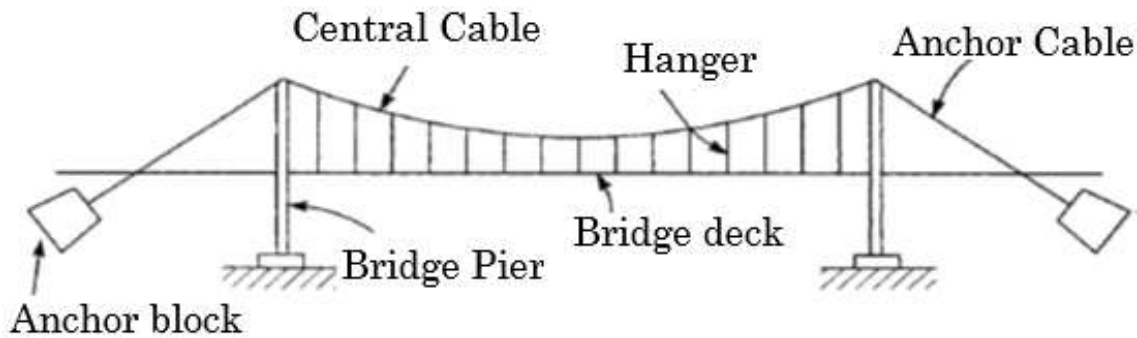


Figure (I.10): Bridge suspended by cables

In the subsequent discussion, we use the notation T_A to represent the tension in the main cable, while T_C denotes the tension in the anchor cable.

I.6.1. The cable passes through a pulley: $T_A = T_C$ (Fig. I.11a)

- Resulting from horizontal forces at the top of the mast (H_T), (β is a value fixed beforehand, $\beta \neq \alpha$)

$$H_T = T_C \cos \alpha - T_A \cos \beta \quad (\text{I.12a})$$

$$H_T = T_C (\cos \alpha - \cos \beta) \quad (\text{I.12b})$$

- Maximum bending moment at the base of the mast ($M_{T(max)}$)

$$M_{T(max)} = H_T h_T = T_C (\cos \alpha - \cos \beta) h_T \quad (\text{I.12c})$$

- Vertical force at the base of the mast (V_T)

$$V_T = T_C (\sin \alpha + \sin \beta) \quad (\text{I.12d})$$

- Weight of the anchor block (ω)

$$\omega = T_A \sin \beta = T_C \sin \beta \quad (\text{I.12e})$$

I.6.2. The cable passes through saddles on roller (Fig. I.11b)

The rollers remain at rest; equilibrium involves zero resulting from horizontal forces at the top of the mast ($H_T = 0$). Consequently, bending moment is zero along the mast.

- The horizontal component tension in the anchor cable is that in the suspension cable, so:

$$T_A \cos \beta = T_C \cos \alpha \quad (\text{I.13a})$$

- The tension in the anchor cable

$$T_A = T_C \left(\frac{\cos \alpha}{\cos \beta} \right) \quad (\text{I.13b})$$

- The vertical force at the base of the mast is:

$$V_T = T_C \sin \alpha + T_A \sin \beta \quad (\text{I.13c})$$

- Weight of the anchor block

$$\omega = T_A \sin \beta \quad (\text{I.13d})$$

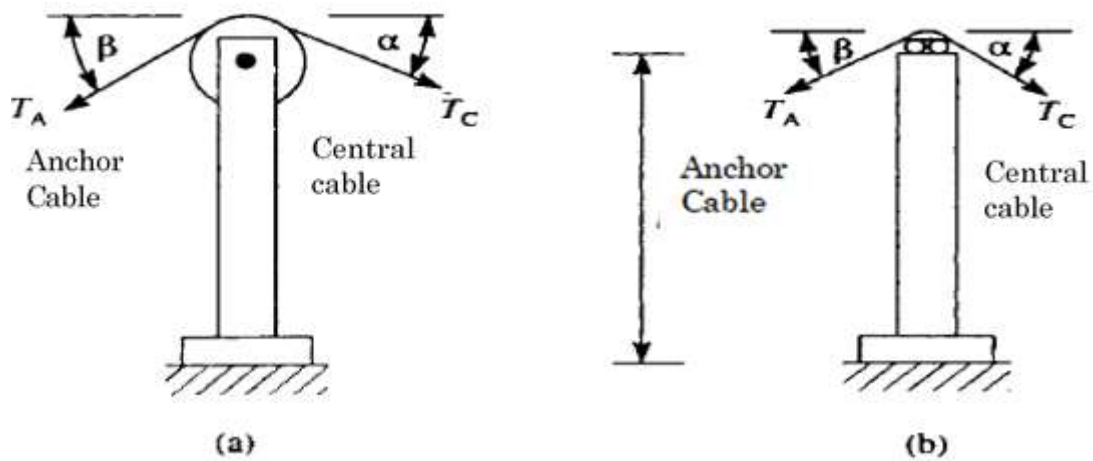
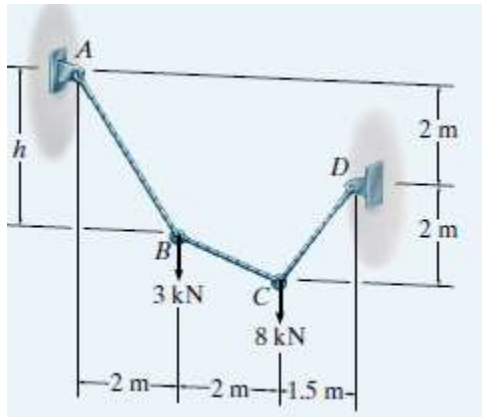


Figure (I.11): (a): Cable passing through a pulley. (b) Cable passing through a roller.

I.7. Problems

Problem 1

Determine the tension in each cable and the dimension h . Determine the inclination of the cable at A and D.

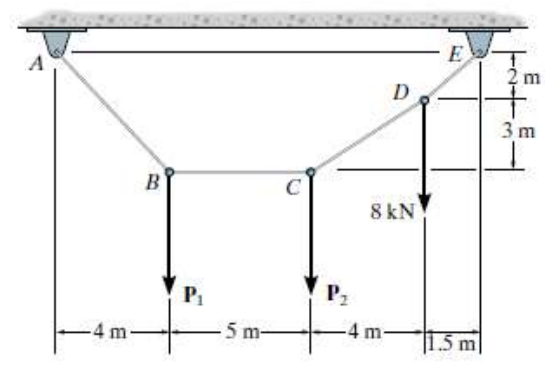


Pb.1

N.A: $H = 4.07 \text{ kN}$, $R_{Ay} = 5.58 \text{ kN}$, $R_{Dy} = 5.42 \text{ kN}$, $T_{AB} = 6.9 \text{ kN}$, $T_{CD} = 6.78 \text{ kN}$, $T_{BC} = 4.81 \text{ kN}$, $h = 2.74 \text{ m}$, $\theta_A = 53.89^\circ$, $\theta_D = 53.09^\circ$.

Problem 2

Determine the forces P_1 , P_2 necessary to maintain the cable in the position shown in the figure, the segment BC remains horizontal.

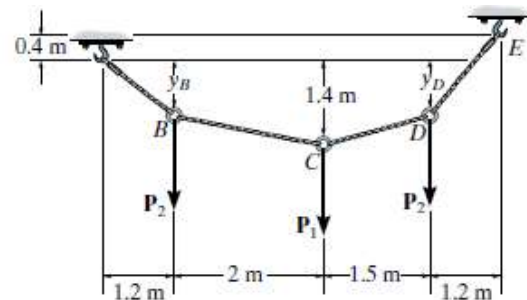


Pb.2

N.A: $P_1 = 17.13 \text{ kN}$, $P_2 = 10.37 \text{ kN}$

Problem 3

One cable supports three loads. Determine the magnitude of P_1 if $P_2 = 3 \text{ kN}$ and $y_b = 0.8 \text{ m}$, calculate the deflection y_D .

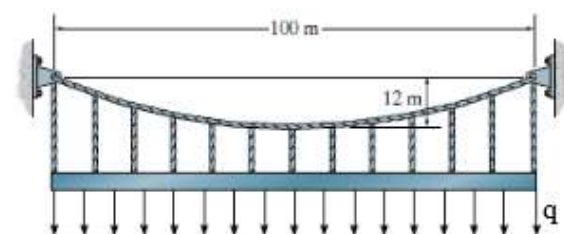


Pb.3

N.A: $P_1 = 6.58 \text{ kN}$, $y_D = 0.644 \text{ m}$

Problem 4

- The suspension cable of a pedestrian bridge breaks when the maximum tension reaches $T_{max} = 5000 \text{ kN}$. Determine the maximum load q uniformly distributed to reach this tension.
- The cable is subjected to a uniformly distributed horizontal load of $q = 30 \text{ kN/m}$. Determine the maximum and minimum tensions in the cable.

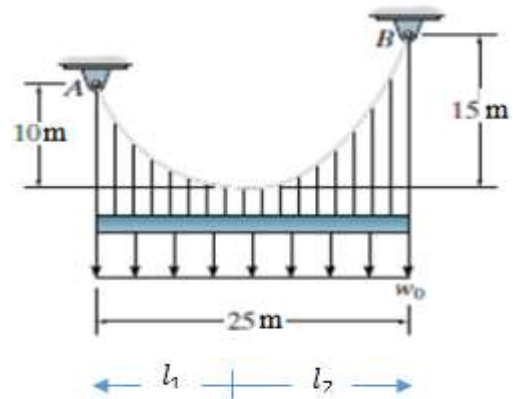


Pb.4

N.A: $q = 43 \text{ kN/m}$, $T_{max} = 3.46 \text{ MN}$, $T_{min} = H = 3.125 \text{ MN}$

Problem 5

A cable is subjected to a total vertical load of $w=450\text{KN}$. Determine the maximum and minimum tension in the cable and the inclination of the cable at the support points.

**Pb.5**

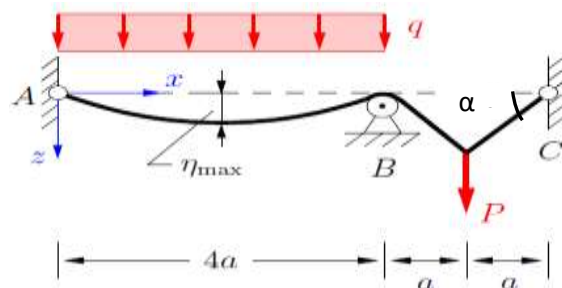
N.A: $l_1 = 11.23\text{m}$, $l_2 = 13.76\text{m}$, $H = 113.5\text{KN}$, $V_A = 202.3\text{KN}$, $V_B = 247.7\text{KN}$, $T_A = 231.82\text{KN}$, $T_B = 272.05\text{KN}$, $\theta_A = 60.7^\circ$, $\theta_B = 65^\circ$

Problem 6

A rope is laid through a frictionless pulley. Determine the load P as a function of ' q , a and α ' for maximum deflection

$$\eta_{\max} = a/10.$$

N.A: $P = 40 * q * a * \sin \alpha$

**Pb.6****Problem 7: Suspended Bridge, anchored on the outside**

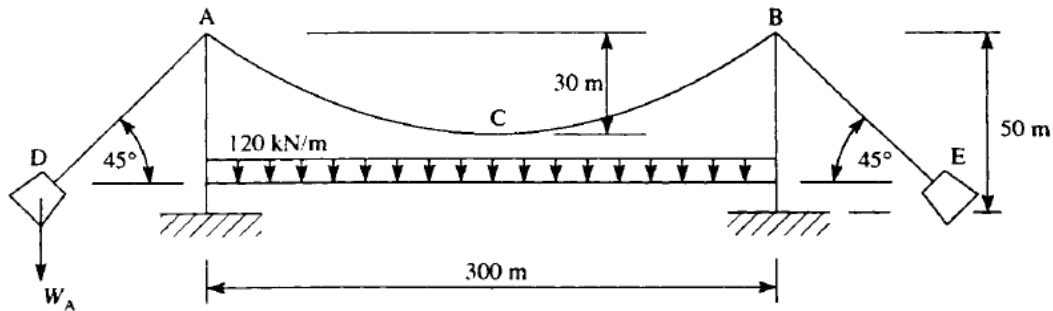
A suspension cable of a suspension bridge supports a load uniform horizontally distributed of 120KN/m , and is anchored in the masses D and E at an angle of 45° . Determine:

1. The maximum tension in the suspension cable, and the minimum diameter of the cable knowing that the allowable stress of the constituent material is 1000N/mm^2 . Calculate the inclination (α) of the suspension cable at supports A and B.
2. If the cable passes through a frictionless pulley at the top of masts A and B, calculate the bending (overturning) moment at the base of the mast, the vertical force at the base of the mast as well as the necessary weight of the anchor block ω_{anc} .
3. If the suspension cable passes through a trolley on a roller, calculate the tension in the anchor cable, the vertical force at the base of the mast as well as the necessary weight of the anchor block ω_{anc} .

N.A: (1): $T_A = T_B = 48.46\text{MN}$, $d_{\min} = 248.4\text{mm}$, $\alpha = 21.8^\circ$

(2): $M_{\text{mât}} = 536,4\text{MN.m}$, $N_{\text{mât}} = 52,27\text{MN}$, $\omega_{anc} = 34\text{MN}$, $F_H = 10.88\text{KN}$, $F_V = 51.86\text{KN}$.

(3): $T = 63.640\text{MN}$, $N_{\text{mât}} = 62.86\text{MN}$, $\omega_{anc} = 45\text{MN}$.



Pb.7: Suspension bridge anchored on the outside.

Problem 8: Dardanelles 1915 Bridge (“Çanakkale 1915” bridge, Turkey)

The Dardanelles Strait suspension bridge in Istanbul, exclusively for motorway use, connects the city of Lapseki to the city of Gelibolu. This strait forms a natural division between Europe and Asia, and it connects the Sea of Marmara to the Aegean and Mediterranean seas. This most important suspended bridge has been operational since 2022, constructed with special steel cables with a resistance of 1960 N/mm^2 and features three lanes of traffic in each direction. The bridge consists of three spans, with a central span measuring $L=2023 \text{ m}$ and two side spans measuring $L=770 \text{ m}$.

Each of the two central cables is supported by two piers with a height of 318 m . Specifically, the pier height up to the deck level is approximately $h_p=236 \text{ m}$. The maximum deflection of the two central cables occurs at the mid-span and measures $f=213.8 \text{ m}$. If the total weight of the bridge deck is 500000 tonnes :

I. Central Cable

- Draw a diagram of this suspension bridge, indicating all the provided data.
- Calculate the weight supported by a cable in Kg/m .
- Write the deflection expression; calculate the maximum tension T_{max} in the cable and the maximum inclination (θ) of the cable.
- Calculate the minimum diameter necessary to reach rupture.
- **Side Cable**

If the suspension cable passes through a trolley on a roller, calculate the tension in the anchor cable near the pile.

Problem 9: Akashi Bridge (Japan)

The Akashi Strait suspension bridge in Japan, exclusively for motorway use, connects the city of Kobe to the city of Awaji and has been operational since 1998. This impressive structure is constructed with special steel cables with a resistance of 1800 N/mm^2 and features three lanes of traffic in each direction. The bridge consists of three spans, with a central span measuring $L=1990.8 \text{ m}$ and two side spans measuring $L=960.3 \text{ m}$.

Each of the two central cables is supported by two piers with a height of 297 m , equivalent to the height of an 80-story tower. Specifically, the pier height up to the deck level is approximately $h_p=216.3 \text{ m}$. The maximum deflection of the two central cables occurs at the mid-span and measures $f=196 \text{ m}$.

Now, if the total weight of the bridge deck is 160,000 tons, we can calculate the uniformly distributed load q (in kN/m) that acts on a cable due to the weight of the deck alone, representing roughly 91% of the total weight supported by the cables.

II. Central Cable

- Write the deflection expression; calculate the maximum tension T_{max} in the cable and the maximum inclination (θ) of the cable.
- Verify the resistance of this cable.
- If in reality $T_{max} = 625\text{MN}$, integrating the load due to road traffic. Calculate the load uniformly distributed on the horizontal q' (KN/m), which really acts on the cables. Continue the rest of the exercise with this load value.

III. Side Cable

Knowing the distance $d=45\text{m}$, deduce the deflection of the side cable at mid-span. Calculate for this side cable:

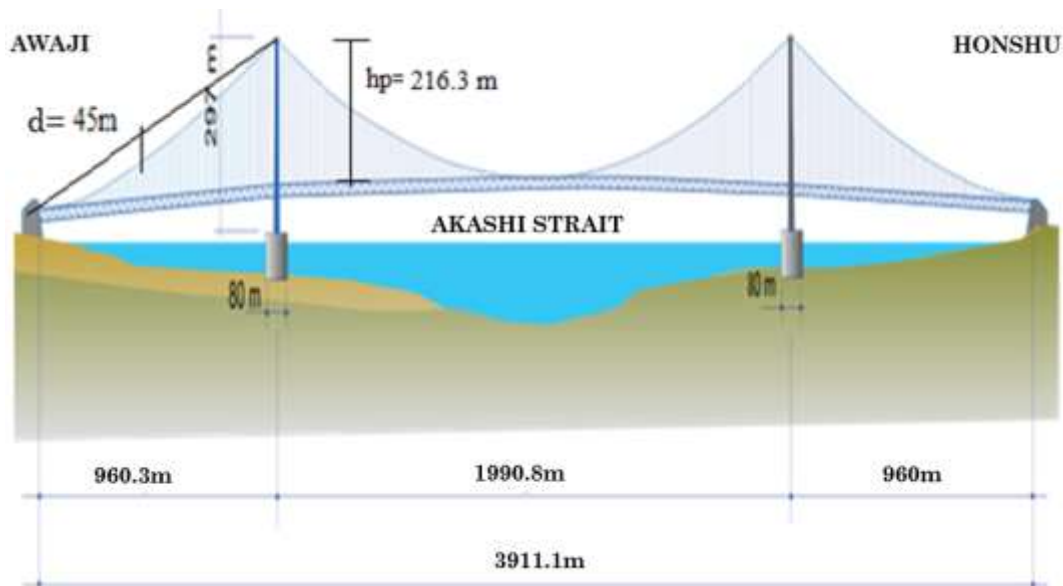
- The inclination of the cable (β) on the bank (anchoring point)
- The inclination (α) of the cable in the vicinity of the pile
- The cable tension on the bank, T_β (anchoring point).
- The cable tension, T_α in the vicinity of the piers

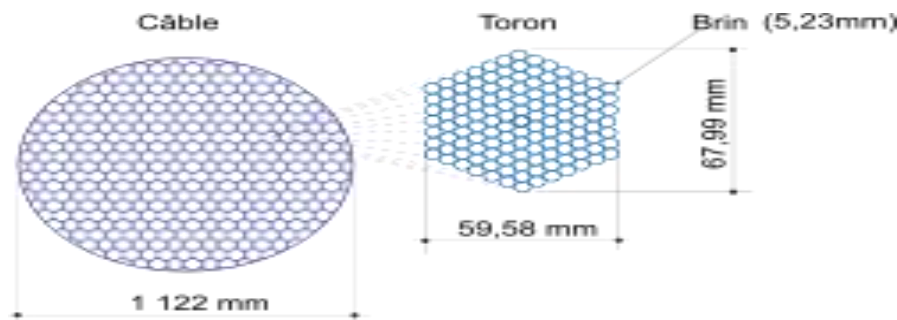
IV. Describe the sealing system used for this bridge. Justify with a calculation.

N.A: $q = 205\text{ KN/m}$.

Central cable: $T_{max} = 556\text{MN}$, $\theta = 21.5^\circ$, $\sigma_{max} = 563.4\text{MPa}$, $q' = 230\text{ KN/m}$.

Side cable: $f = 63.15\text{m}$, $\beta = 2.16^\circ$, $\alpha = 22.31^\circ$, $T_\beta \cong H = 490.29\text{ MN}$, $T_\alpha = 530.04\text{MN}$.





Pb. 9: On top: Akashi Bridge with 3 spans (2 banks and 1 central), it is self-anchored on the bank. On bottom: Cross-section of the suspension cable.

Problem 10: Golden Gate Bridge (San Francisco)

Consider the Golden Gate suspension bridge, in operation since 1937. Built with steel cables with a strength of 1100 N/mm^2 and a diameter of $d=92\text{cm}$. It has (03) spans, a central one with a length $L=1280\text{m}$ and two side spans with a length $L'=343\text{m}$. Each of the two central cables is supported by two piers with a total height of 227m , which means a pier height of approximately $h_p=152\text{m}$ up to the level of the deck

If one principal cable is subjected to uniform horizontally distributed load of $q=165 \text{ KN/m}$.

I. Central Cable

- Calculate the deflection 'h' if the minimal in the central cable is $H=237 \text{ MN}$.
- The maximum tension in T_{max} (MN) and the maximum inclination (θ) of the cable (near the piers).
- What do you think about the resistance of the cables? Justify with a calculation.

II. Side Cable:

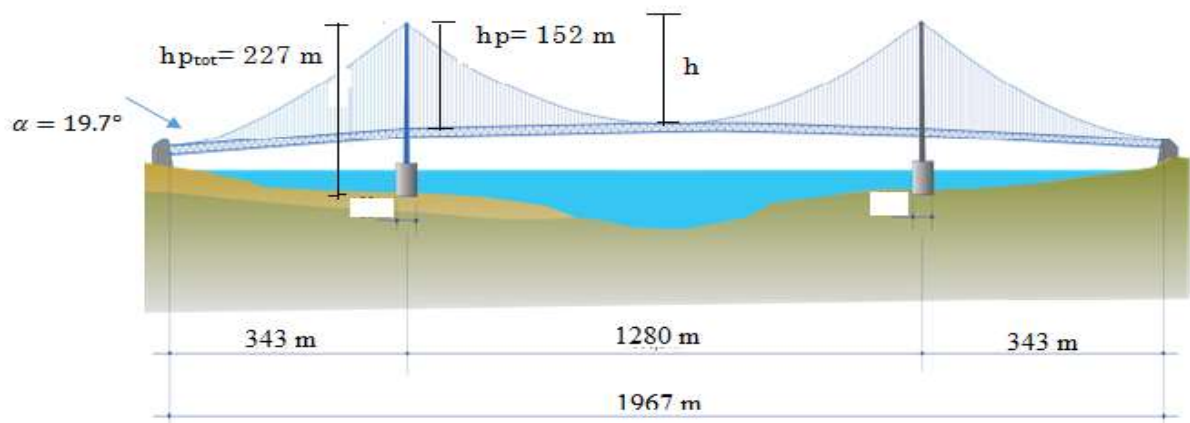
If the inclination of the side cable on the bank (anchoring point) is $\alpha = 19.7^\circ$:

- Calculate the inclination (β) of this cable near the pile.
- Calculate the tension of this cable on the bank T_α .
- Calculate the inking mass in (MN). If the actual inking mass is 534MN , do you think there may be a risk of the cables lifting or not?

III. Describe the sealing system used for this bridge. Justify with a calculation.

NA: .I. Central cable: $h = 143\text{m}$, $T_{max} = 259.4\text{MN}$, $\theta = 24^\circ$, $\sigma_{max} = 390.2\text{MPa}$.

II. Side cable: $\beta = 27.83^\circ$, $T_\alpha = 353.10 \text{ MN}$, $\omega_{anc}=119 \text{ MN}$



Pb.10: Self-anchored 'Golden Gate' bridge (San Francisco, USA)' having 3 spans (2 on the banks and 1 central), it is self-anchored on the bank.